

Course Syllabus: Probability and Statistics Foundations

Scope: Lectures 1-4 | Uncertainty, conditional probability, random variables, distributions, and modelling

Course purpose

This short foundation course develops a disciplined way to describe uncertainty, update beliefs using evidence, represent outcomes with random variables, and select simple probability models. Probability and statistics are taught as intertwined tools: probability expresses assumptions about uncertain mechanisms, while data are used to check, estimate, and revise those assumptions.

Learning outcomes

By the end of the course, students should be able to:

1. Define experiments, outcomes, sample spaces, and events, and apply basic set operations.
2. Use probability rules, counting methods, conditional probability, the multiplication rule, total probability, and Bayes rule.
3. Distinguish independence from mutual exclusivity and test independence from probabilities or contingency tables.
4. Define discrete and continuous random variables, PMFs/PDFs, CDFs, expectation, variance, medians, and quantiles.
5. Recognize and use simple Bernoulli, binomial, discrete uniform, and continuous uniform models.
6. State modelling assumptions, judge their plausibility, interpret results in context, and verify calculations.

Prerequisites

Basic algebra, manipulation of fractions and powers, reading graphs, and summation notation. Calculus is helpful for continuous distributions but is not required for the core uniform-distribution examples.

Course structure

Unit	Main ideas	Student evidence
1. Uncertainty	Experiments, sample spaces, events, set notation, probability axioms, complements, unions, intersections, counting and interpretation.	Construct a sample space; translate a verbal question into events; check whether an answer is plausible.
2. Conditioning	Conditional probability, multiplication rule, total probability, Bayes rule, permutations/combinations, independence.	Solve forward and reverse conditional questions; explain the base-rate effect.
3. Random variables	Discrete/continuous variables, PMFs/PDFs, CDFs, expectation, variance, descriptive summaries.	Build a distribution, calculate probabilities, and interpret expected value and spread.
4. Modelling	Median and quantiles; families of discrete and continuous models; assumptions, fit, limitations, and verification.	Select a model, justify parameters, critique assumptions, and communicate limitations.

Teaching and learning activities

Before class: short concept video or reading, followed by two diagnostic questions. During class: worked examples, prediction questions, peer explanation, and short individual practice. After class: mixed exercises and one reflection prompt asking which assumption was most important.

Recommended sequence

Session	Focus	Suggested activity
1	Language of uncertainty	Build sample spaces for dice, cards, and a measurement process; compare intuitive and formal answers.
2	Counting and probability rules	Queue/chair permutations; combinations; Venn-diagram checks.
3	Conditional probability	Tree diagram and table representations; medical-test or machine-quality Bayes example.
4	Random variables and CDFs	Map outcomes to numbers; construct a PMF; read probabilities from a CDF.
5	Continuous distributions and quantiles	Area under a uniform density; locate median and quantiles on the CDF.
6	Modelling and integration	Select and critique a model for alarms, defects, arrivals, or measurement error.

Assessment plan

Formative (ungraded): pre-class checks, one-minute papers, and worked-example completion.

Summative: 20% weekly exercises, 20% modelling memo, and 60% closed-book exam. The exam should balance calculation, representation, interpretation, and critique rather than reward formula recall alone.

Suggested grade weighting by capability

Formal setup	30%	Events, assumptions, notation, support, and model parameters.
Calculation	35%	Correct use of rules, algebra, and numerical accuracy.
Interpretation	20%	Meaning of probabilities, expectations, quantiles, and conditional results.
Verification/critique	15%	Checks, limitations, dependence, and consequences of assumptions.

Course policies

Calculator and software: permitted for arithmetic and graphing after the probability model and formula are stated. Collaboration: discussion is encouraged on formative work, but submitted solutions must be written independently. AI tools: may be used for practice explanations or feedback when allowed by the instructor; students remain responsible for verifying formulas, assumptions, and final reasoning. Late work: use the institution's standard policy, with reasonable accommodations documented in advance.

Accessibility and inclusion

Provide formula sheets, readable notation, alternative text for diagrams, and extra processing time where approved. Use contexts from engineering, environment, health, and everyday decisions without assuming specialized background knowledge. Avoid assessing language complexity when the learning target is probability reasoning.

Required habits of mind

Every substantial solution should answer four questions: What is known? What is being asked? Which model or rule connects them? Does the result make sense? Students should label events and random variables before calculating and should interpret the final number in the original context.

Core formula and concept list

Complement and addition rules; conditional probability; multiplication rule; law of total probability; Bayes rule; permutations and combinations; independence; PMF/PDF/CDF; expectation and variance; median and quantiles; Bernoulli, binomial, discrete uniform, and continuous uniform models.

Completion standard

A student has met the course standard when they can solve unfamiliar but structurally similar problems, explain why the selected method applies, and identify at least one meaningful assumption or limitation.